AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 15 · STUDENT LESSON

Save New Examples for Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 13: fitting a rule differs from checking unfamiliar examples.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Separate examples used to choose a model from examples reserved for its final evaluation. Try the recall before opening notes.

Your learning target

I can separate examples used to choose a model from examples reserved for its final evaluation.

I can record a fixed model and its **prediction**s before revealing comparison labels.

I can explain how test feedback changes the **evidence** available for a revised model.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A fair test uses examples not used to create the decision **process**.



Your target

LEARNING TARGET Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Learner name:

Readiness for the connected explanation

A height passes the **input** contract and the model predicts tall. What is still needed to check that **prediction**? If its comparison label changes your threshold choice, what role has that example played?

This week's end goal

By the end, I can **freeze** a model before labels are revealed and report a bounded test result.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Save New Examples for Testing case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A rule chosen to fit examples may work differently on new cases. To investigate that difference, reserve a set of examples before development. Use the **development examples** to choose the model.

Keep the final test examples separate until the model is fixed, then evaluate that recorded model.

In this paper activity, a teacher keeps the reserved examples in an envelope. Record the **input** contract, feature, category rule and model version before opening it.

Once the **input** values are revealed, predict using the frozen model before revealing their comparison labels. This prevents copying the labels into the **prediction**s.

The worked example below describes a completed experiment, so its test labels are visible for discussion. It is not an unseen test for you.

In the workbook experiment, the teacher will supply a separate set of comparison labels after you have saved your **prediction**s. All data are imaginary classroom records.

STUDENT LESSON · 3 OF 18

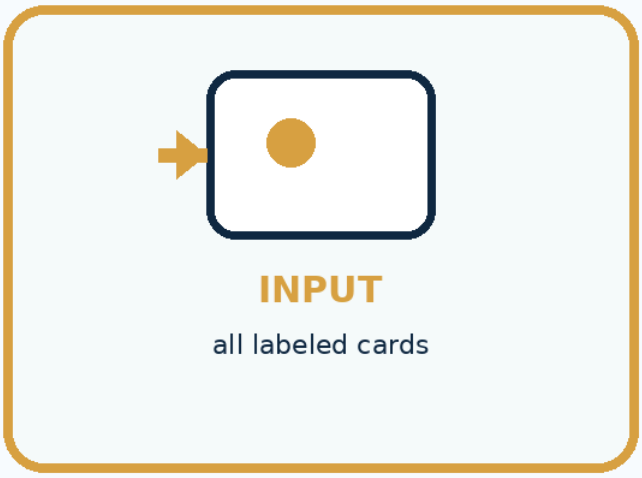
See how it works

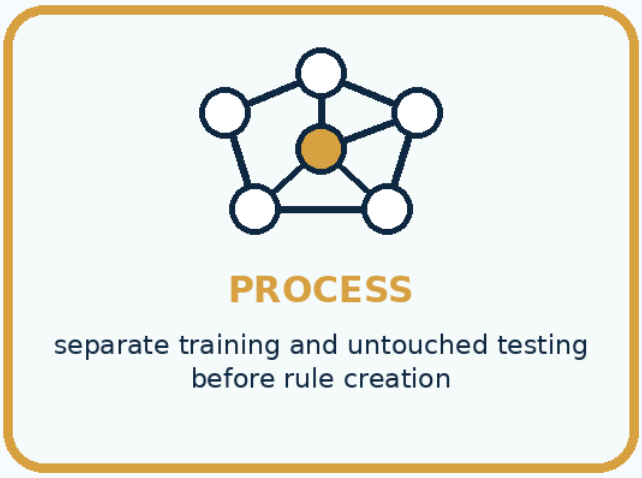
> Follow the mechanism for Save New Examples for Testing. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT all labeled cards

PROCESS separate **training** and untouched testing before rule creation

OUTPUT honest performance **evidence** on new cards

Key idea

CORE IDEA A fair test uses examples not used to create the decision **process**.

How the mechanism works

UNDER THE HOOD Evaluation on **training** data is optimistically biased. A **holdout test set** estimates behavior on unseen examples only if it remains untouched during design.

Repeatedly tuning to test results leaks information and turns the test into **training**.

Draw a fresh example with all three stages:

A closer explanation

Different roles for development and final testing

EVIDENCE AVAILABLE	ALLOWED STEP	RECORD TO KEEP
Development examples	Fit and select a model	Choice and its reasons
Reserved inputs; labels hidden	Use the frozen model	Predictions and version
Reserved labels revealed	Compare without revising	Results for that version

Read the rows in order. Information from the final comparison must not travel back into model selection while the result is still described as an independent final test of that model.

In the completed experiment, development towers at heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Candidates predict tall at or above 5 or 6 cm.

Both fit all four labels; the stated smaller-threshold tie rule selects 5. The team **freeze**s model TowerA with threshold 5.

The reserved **input**s are different records at heights 3, 5, 7 and 9 cm. TowerA accepts whole cm heights from 1 through 10. It predicts short, tall, tall and tall.

These **prediction**s are saved before the reserved labels are opened. The comparison below records what happened, including the failure.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

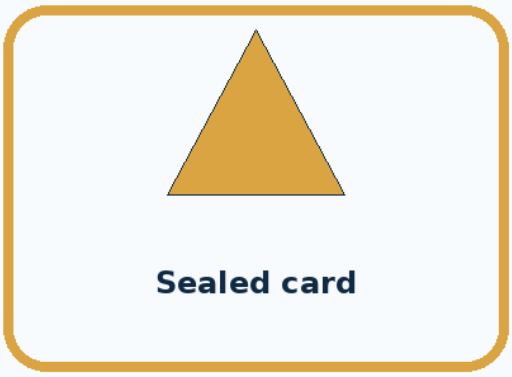
> Compare the supplied cases for Save New Examples for Testing. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Card used to write rule Clue: belongs in learn pile. Meaning: **training evidence**.

The case to investigate

CASE 2 — Sealed card Clue: unseen until final run. Meaning: test **evidence**.

The case to investigate

CASE 3 — Peek then revise Clue: test influenced **process**. Meaning: leakage.

The case to investigate

CASE 4 — Fresh local card Clue: new after design. Meaning: strong transfer check.

Keep this boundary clear

BOUNDARY A **test set** stops being independent when its answers guide revisions.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: After feedback, change the boundary to 5.

Predicted result

Prediction: The revised rule matches all four known labels but needs a different final test.

Why the result follows

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Save New Examples for Testing to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
training set	examples used to create or adjust a process	
test set	untouched examples used for final checking	

Word	Meaning	My example
holdout	data deliberately kept separate	
data leakage	test information influences design	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR Testing on familiar cards is fair because there are more correct answers.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Development examples	Examples available for fitting, comparing or revising a model
Held-out test set	Examples reserved from development for evaluating a fixed model
Freeze	Record a model and stop changing it for a particular evaluation
Test leakage	Information from a reserved test influencing model development
Evaluation record	A record of the tested model, predictions, comparison labels and results

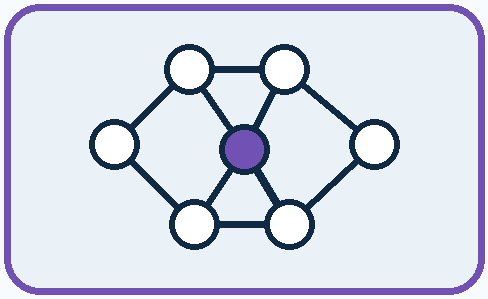
Follow a worked example

> Follow the worked trace for Save New Examples for Testing, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

Step 1

STEP 1 — OBSERVE all labeled cards

Step 2

STEP 2 — TRACE separate **training** and untouched testing before rule creation

Step 3

STEP 3 — CONCLUDE honest performance **evidence** on new cards

Model answer

MODEL REASONING Randomly or deliberately reserve representative cards first. Create the decision **process** using only the learning pile.

Open the test pile once for **evidence**, record every result, and do not call the revised **process** tested until a new **holdout** exists.

A claim to avoid

DO NOT OVERCLAIM A **test set** stops being independent when its answers guide revisions.

What new **evidence** could change this answer?

A closer explanation

Reserved height	Saved TowerA prediction	Revealed comparison label
3 cm	Short	Short
5 cm	Tall	Short
7 cm	Tall	Tall
9 cm	Tall	Tall

TowerA agrees on three of these four reserved examples and disagrees at 5 cm. That is **evidence** about this fixed model on this small set. It does not prove a success rate for every future tower.

The labels, range of examples and way records were selected also matter.

Worked explanation · Save New Examples for Testing

After seeing the 5 cm label, the team changes the threshold to 6 and names the revision TowerB. TowerB agrees on all four of these now-seen examples.

The change may be useful, but those examples have influenced its development. Four agreements on them are not an independent final test result for TowerB. Keep TowerA's original three-of-four record.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Save New Examples for Testing

A fictional envelope exercise evaluates a fixed pencil-length sorter.

The facts we will use

Supplied facts

Freeze PencilP: long at least 6 cm; short below 6.

Supplied facts

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

AI ACADEMY / AI-1/15

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/15

Save New Examples for Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze

Write PencilP and its boundary before opening the label envelope.



2. Predict

For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references

Envelope labels are short,long,long,long.



4. Compare

Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING

After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze	Write PencilP and its boundary before opening the label envelope.
2. Predict	For 3,5,6,9 cm predict short,short,long,long.
3. Reveal references	Envelope labels are short,long,long,long.
4. Compare	Three of four agree; the disagreement is at 5 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

What this example does not establish

Keep the stated limit

Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Keep the stated limit

Read the labelled picture: Does 4/4 on the replay replace the original 3/4 result? Explain.

Keep the stated limit

Change one thing: After feedback, change the boundary to 5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Save New Examples for Testing claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Card used to write rule	belongs in learn pile	training evidence

Case	Evidence	What it shows
Sealed card	unseen until final run	test evidence
Peek then revise	test influenced process	leakage
Fresh local card	new after design	strong transfer check

Repair two tempting claims

Challenge claim A

CLAIM A Testing on familiar cards is fair because there are more correct answers.

Repair:

Challenge claim B

CLAIM B Looking at test cards without labels cannot affect design.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Keeping a set unseen is necessary for this kind of independent check, but secrecy alone does not make it a good test. Four near-identical easy cases provide little **evidence** about varied cases.

Copies of development records can also make a reserved set misleading even if their file names differ.

Check the claim against the facts

Define what future cases the model is meant to handle and choose evaluation examples that address that use. Keep related or copied records together when separating sets.

Record invalid **input**s and no **prediction** outcomes separately instead of quietly dropping them from the report. Later weeks examine coverage and performance comparisons in more detail.

A team tries ten thresholds, reads the reserved test score for each, and reports only the best score as its final unseen test. Explain the problem and give an honest description of what it has measured

---

STUDENT LESSON · 8 OF 18

Practise together

Practise Save New Examples for Testing with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A fair test uses examples not used to create the decision **process**.

Use a fresh example to explain **training set**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Looking at test cards without labels cannot affect design.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Improvement after a test is allowed. State that the old test informed the revision, preserve both versions, and reserve fresh appropriate examples to evaluate the revised model independently.

If none are available, report that limitation instead of inventing a fresh test result.

> **Write two result statements: one for TowerA on the original reserved set, and one for TowerB after the revision. Explain what must happen before a new independent evaluation of TowerB. Would shuffling the four old records make them unseen again**

A test can lose independence even when you never copy its labels into a **training** table. Repeatedly trying thresholds and choosing the one with the best test score also uses test information to choose the model.

Separate development comparisons from the final check.

Practise with less help: Preserve the original result

Use workbook W2 for the connected example “Keeping final tests unseen”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Keeping final tests unseen. Proposed explanation: “A new file makes old records unseen”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does 4/4 on the replay replace the original 3/4 result? Explain.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

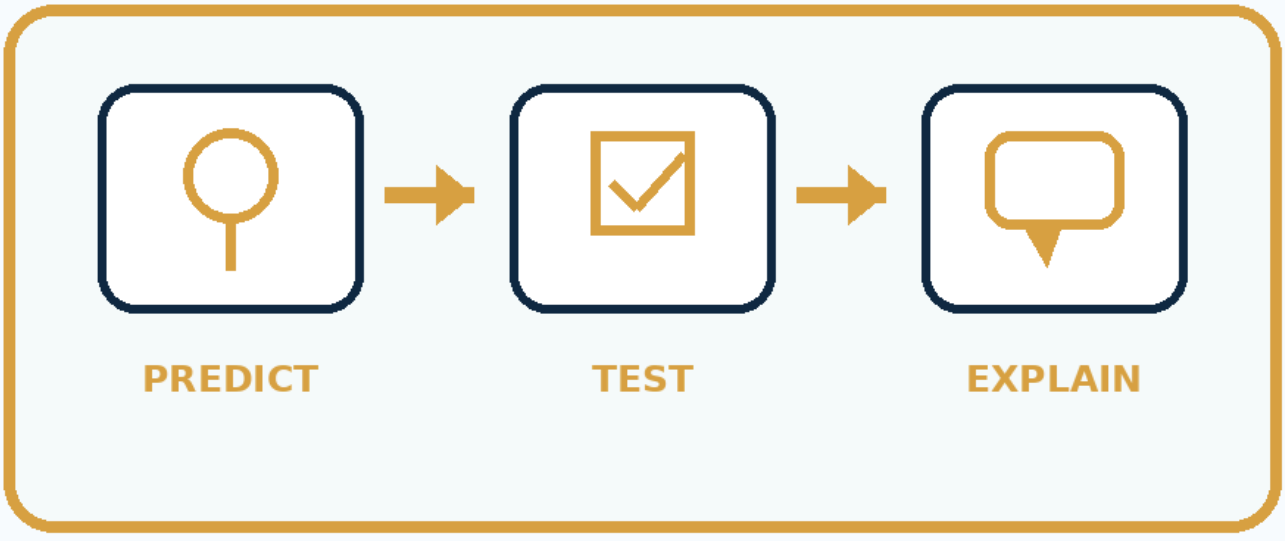
Run the investigation

> Investigate Save New Examples for Testing. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Learn/Test Envelope Split

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Split card data into learn and test piles before sorting.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

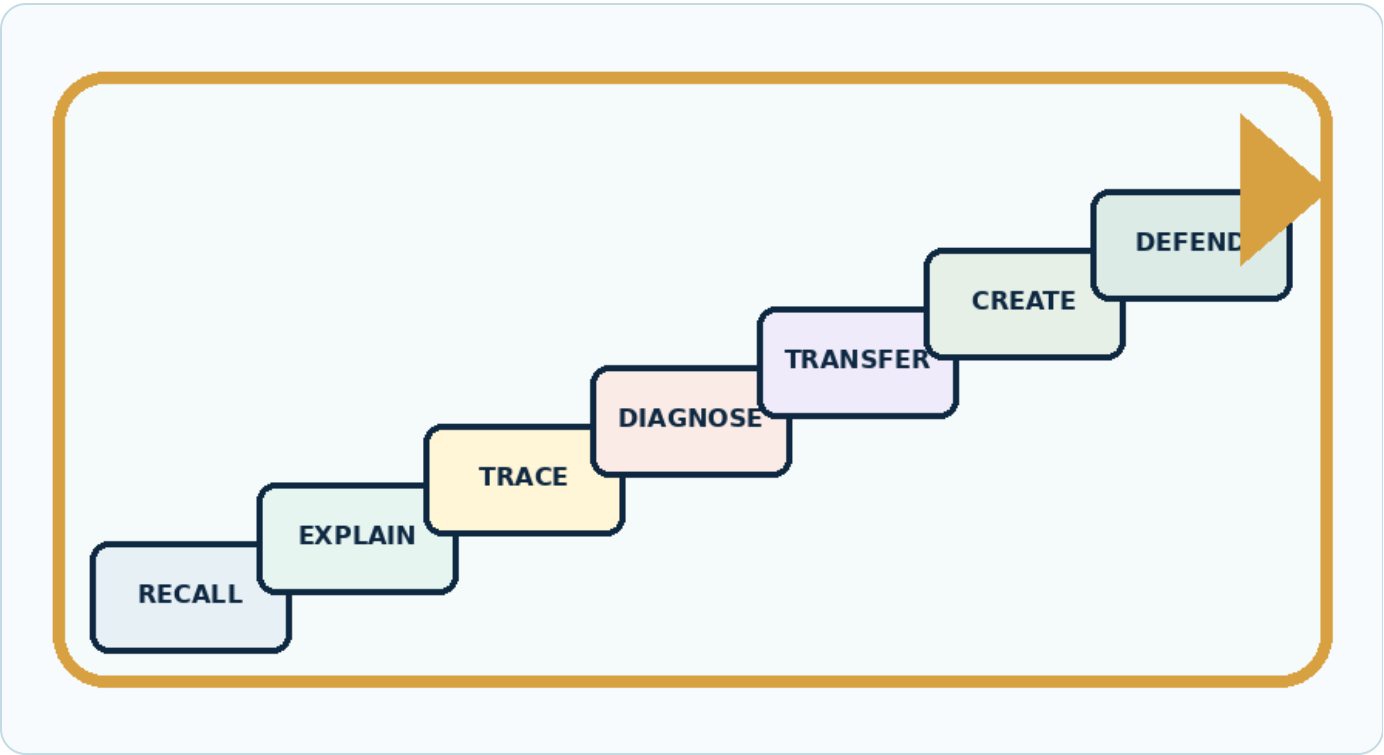
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Save New Examples for Testing. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **training set**.

EXPLAIN A fair test uses examples not used to create the decision **process**.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Testing on familiar cards is fair because there are more correct answers.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

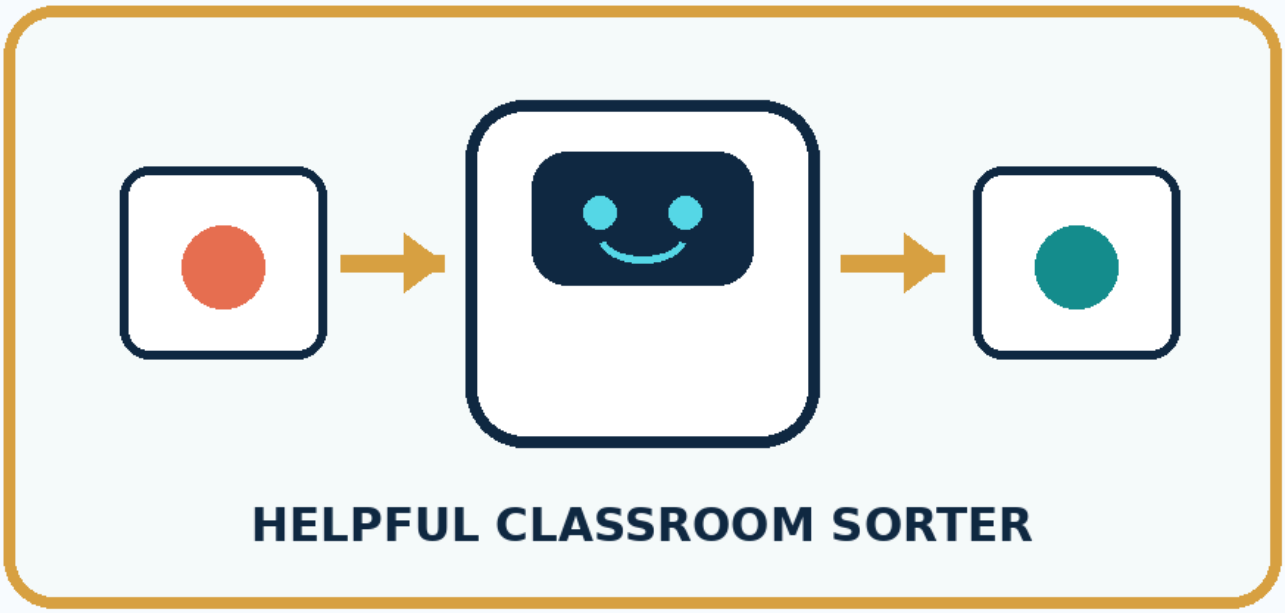
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



> THIS WEEK'S CHECKPOINT Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

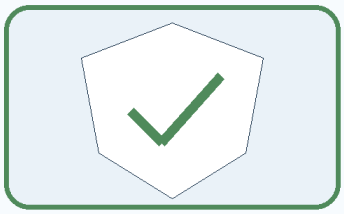
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A **test set** stops being independent when its answers guide revisions.

My project decision:

Evidence for your project

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Split card data before building the sorter, explain why the test pile stays

sealed, and report performance on it.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Randomly or deliberately reserve representative cards first. Create the decision **process** using only the learning pile.

Open the test pile once for **evidence**, record every result, and do not call the revised **process** tested until a new **holdout** exists.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Boundaries and 'Not Sure'.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Save New Examples for Testing case for supported practice. Repeated

> examples do not become fresh assessments.

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Cards A–F are used for practice. G and H remain sealed.

The question

> Which can provide a fresh test, and under what condition?

Worked explanation

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep the final cards unseen

Evidence or stage	Value or result
Practice A–F	6
Sealed test G–H	2

Cards A–F are used for practice. G and H remain sealed.

> Which can provide a fresh test, and under what condition?

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Run a separate ribbon experiment

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Save New Examples for Testing, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional envelope exercise evaluates a fixed pencil-length sorter.

By the end, I can	How I will show it
Separate examples used to choose a model from examples reserved for its final evaluation	Explain the deciding step, show a trace or test, and state one limit.
Record a fixed model and its predictions before revealing comparison labels	Explain the deciding step, show a trace or test, and state one limit.
Explain how test feedback changes the evidence available for a revised model	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Save New Examples for Testing without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate examples used to choose a model from examples reserved for its final evaluation

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four agree; the disagreement is at 5 cm.

1. Reveal references

2. Compare

3. **Freeze**

4. Predict

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare Three of four agree; the disagreement is at 5 cm. The **prediction** record must exist before the labels are seen.

PencilP scores 3/4=75% agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: After feedback, change the boundary to 5. Which conclusion is supported by the supplied facts?

1. The revised rule matches all four known labels but needs a different final test.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The revised rule matches all four known labels but needs a different final test.

> These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Repair a missing building block

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

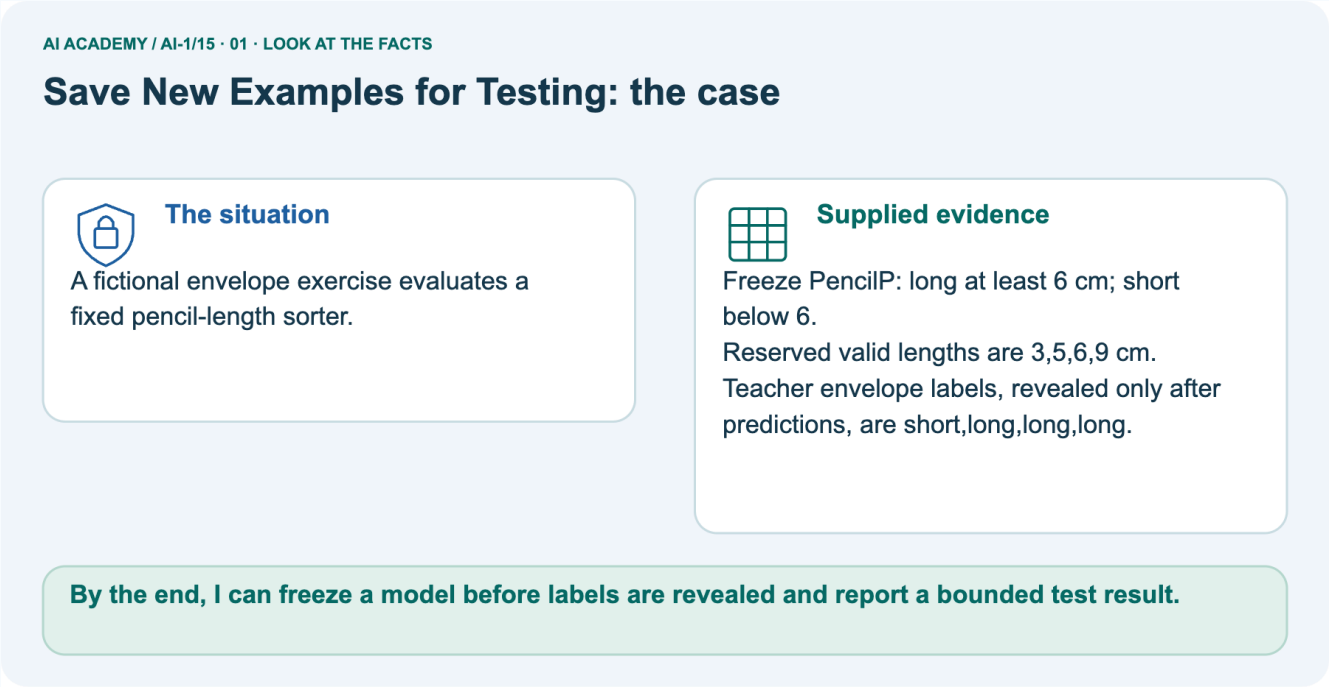
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Save New Examples for Testing graphic using its labels, deciding evidence and relevant limitation.

01. Save New Examples for Testing: the case



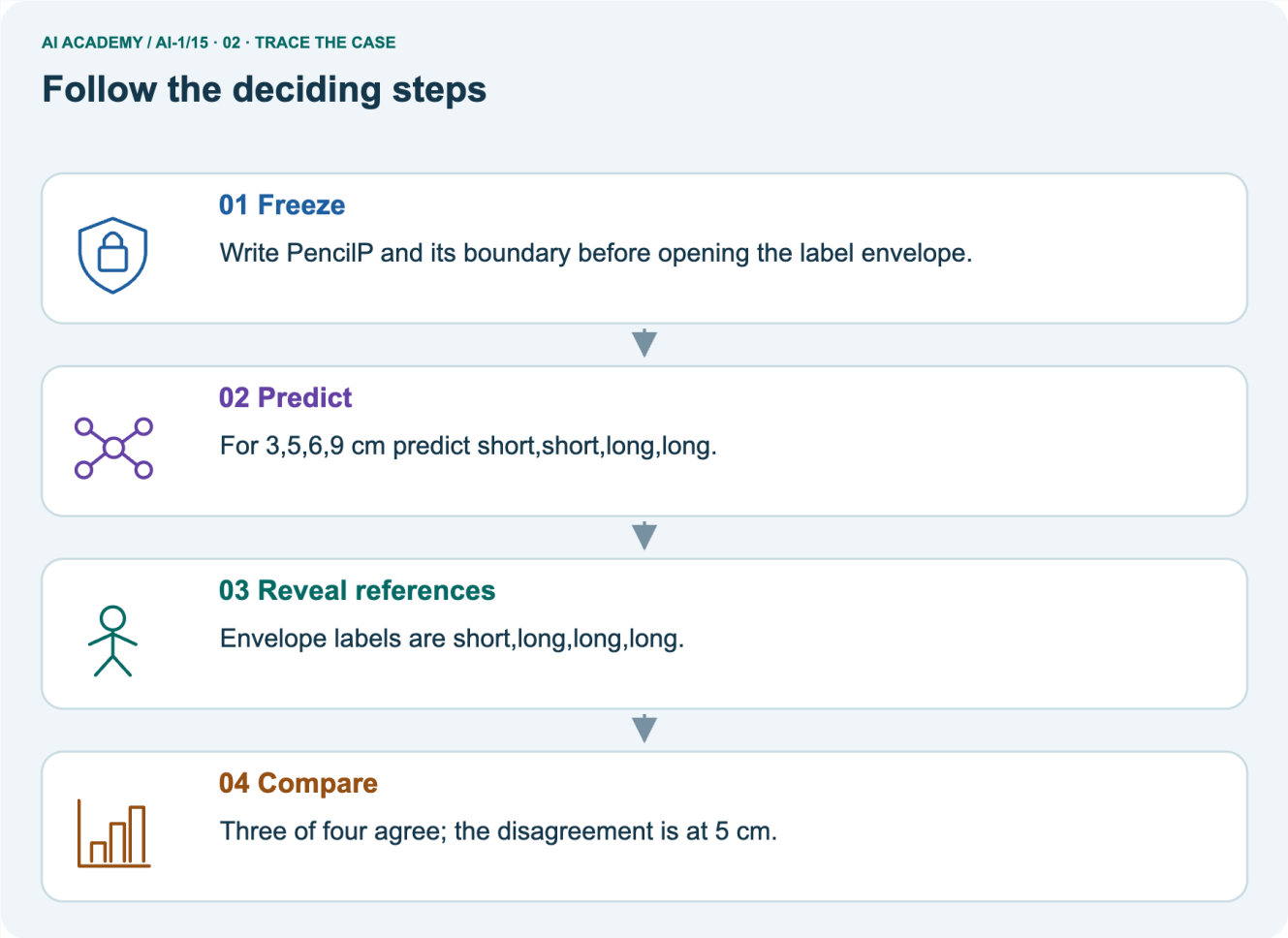
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional envelope exercise evaluates a fixed pencil-length sorter. Supplied facts: **Freeze** PencilP: long at least 6 cm; short below 6. Reserved valid lengths are 3,5,6,9 cm.

Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

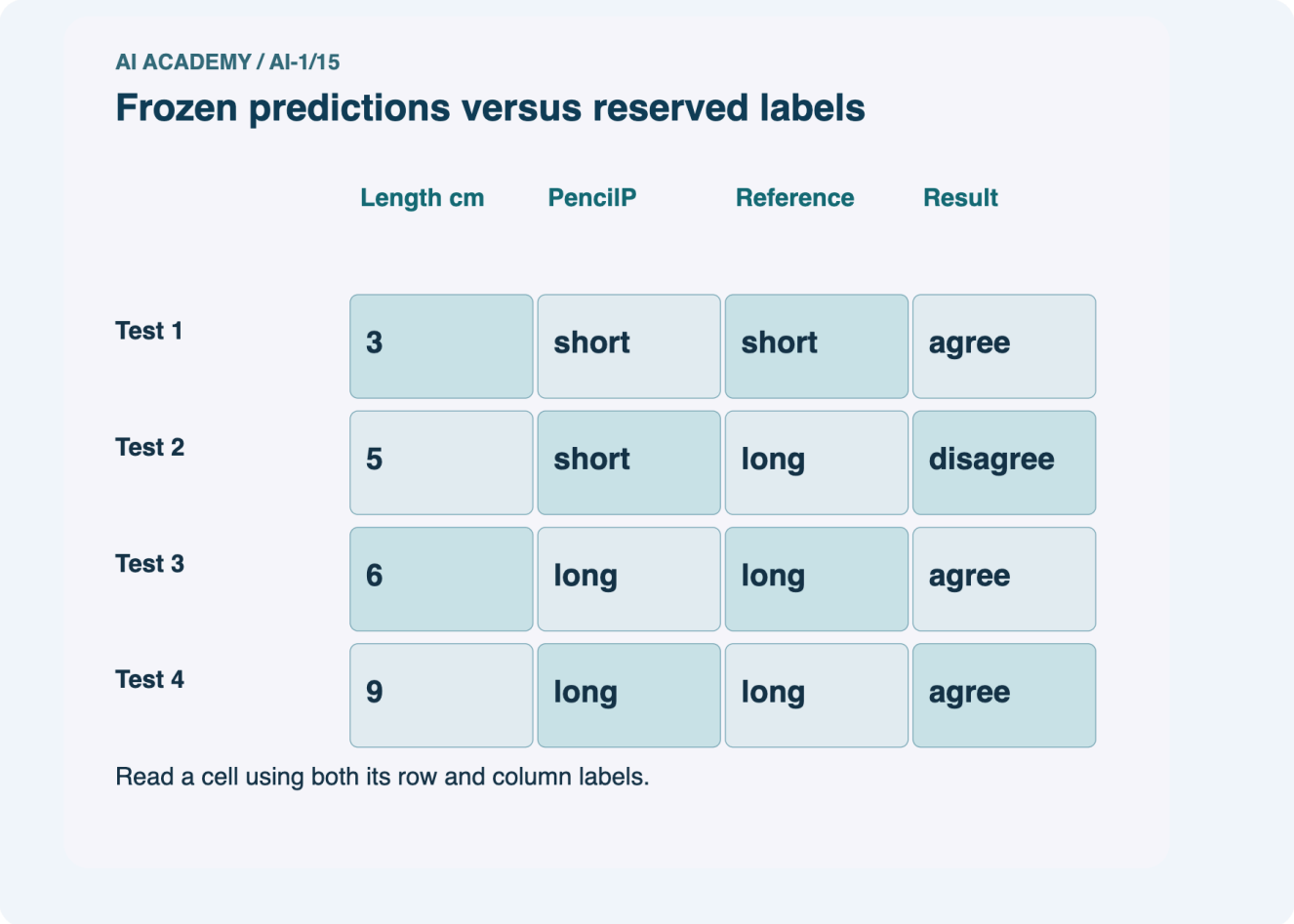


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze: Write PencilP and its boundary before opening the label envelope. → Predict: For 3,5,6,9 cm predict short,short,long,long. → Reveal references: Envelope labels are short,long,long,long. → Compare: Three of four agree; the disagreement is at 5 cm.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Frozen predictions versus reserved labels



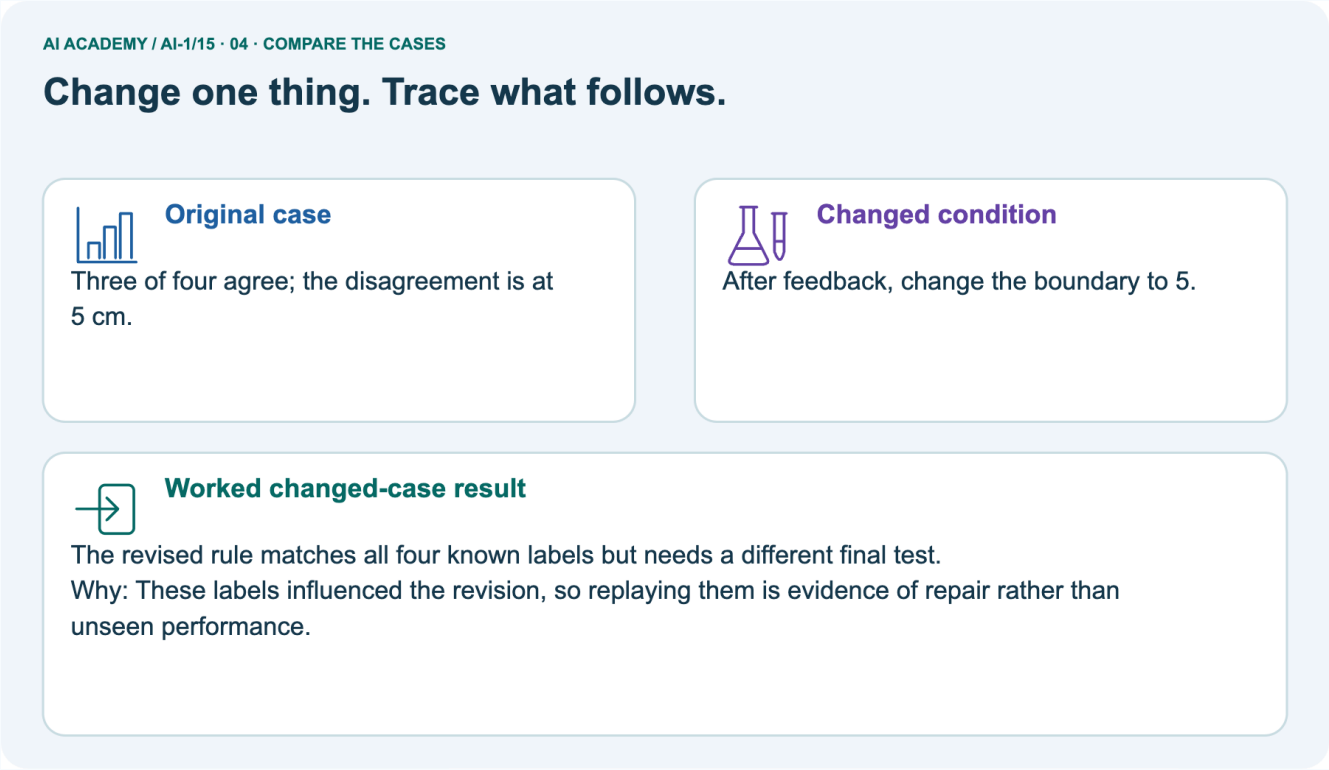
Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen **prediction**s versus reserved labels. Test 1: Length cm = 3, PencilP = short, Reference = short, Result = agree. Test 2: Length cm = 5, PencilP = short, Reference = long, Result = disagree.

Test 3: Length cm = 6, PencilP = long, Reference = long, Result = agree. Test 4: Length cm = 9, PencilP = long, Reference = long, Result = agree.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



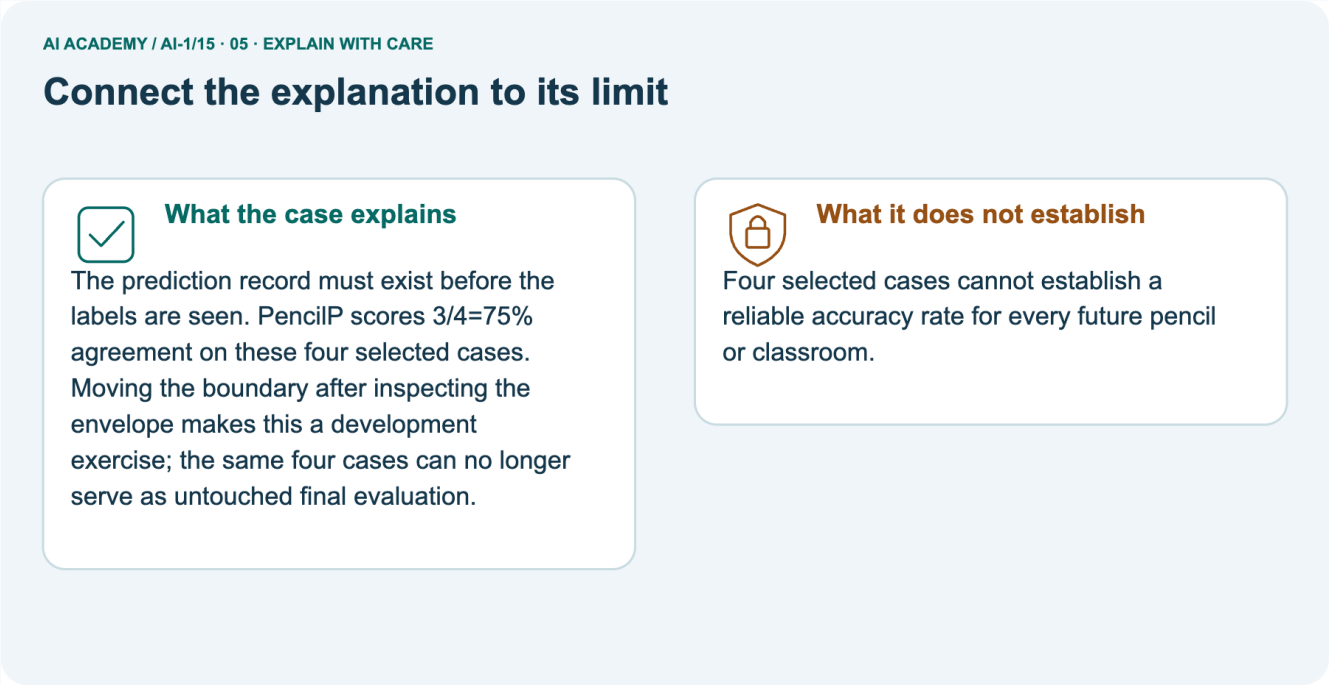
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four agree; the disagreement is at 5 cm. Changed condition: After feedback, change the boundary to 5. Expected result: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

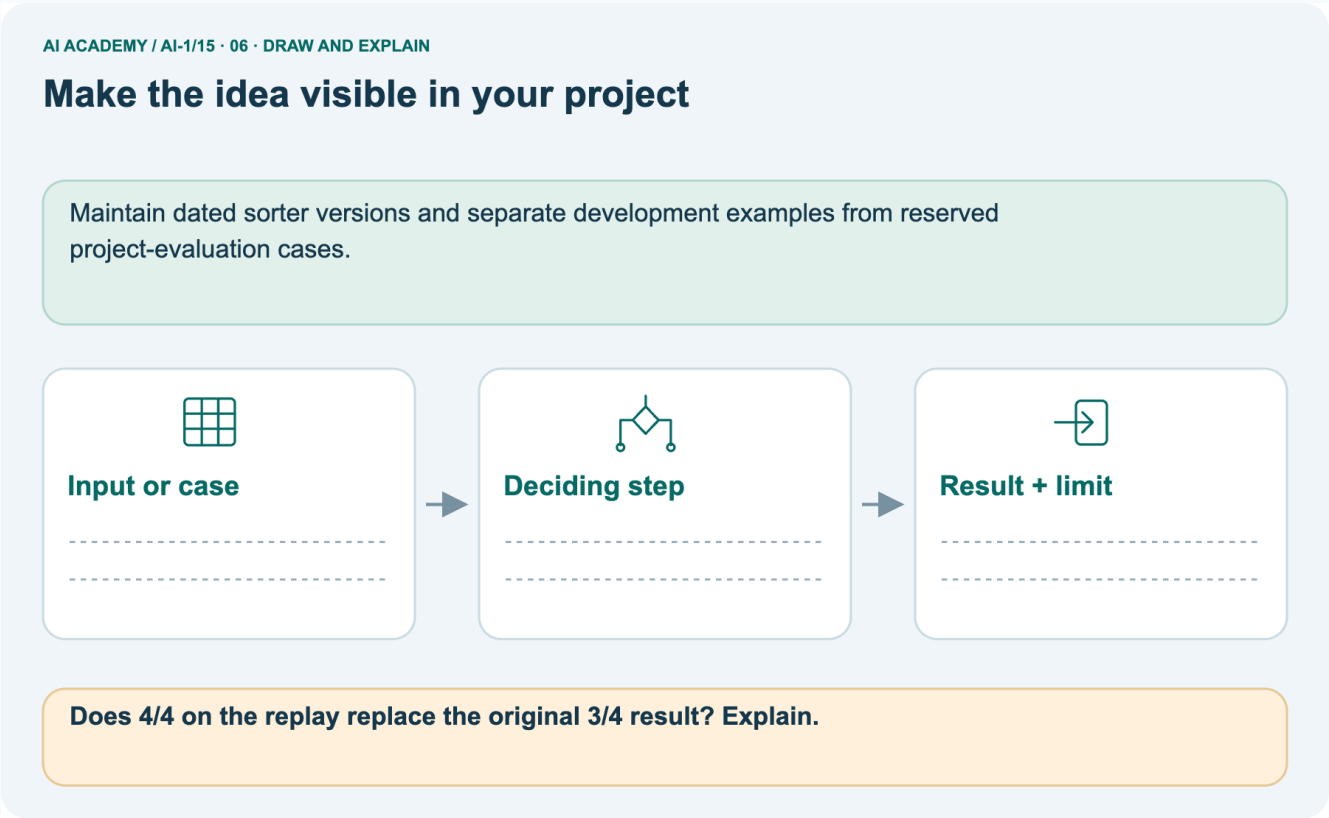
The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Limit: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases. Draw your **input**, deciding step, result and limit.

Supported question: Does 4/4 on the replay replace the original 3/4 result? Explain.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.